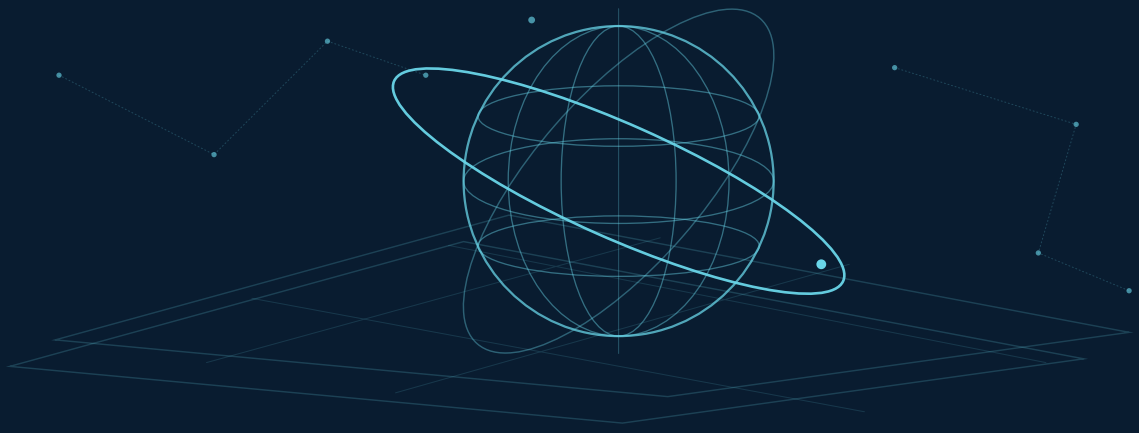


SPECIFICATION

Minimal request register

A complete small Atlas instance with one entity, validation, tool, case, requirement set, and web boundary.

ATLAS-EX-MINIMAL

Revision 2 / draft / Stable-v1.0.0

Document	ATLAS-EX-MINIMAL	Revision	2
Status	draft	Language	en
Atlas version	Stable-v1.0.0	Profile version	Stable-v1.0.0
Translator	1.0.0	Created	2026-10-25
Updated	2026-10-25		

Authors

Identity / name	Contract	Description / details
Douglas de Oliveira Alkimim		Role: Specification author

Keywords: minimal; request; validation; boundary

Context, actors, and scope

The request register retains requests identified by code. It is a declarative specification; no business application is built here.

The scope covers submitting one request. Retrieval, deletion, pricing, stock, authentication implementation, and multi-tenant operation are outside this example.

> The tool is deliberately abstract: it states a persistence contract without choosing storage technology.

This is a synthetic specification example. Source SRC-REQUEST-BRIEF is a teaching baseline, not evidence of stakeholder approval or a claimed Needs-to-L1 derivation.

Actor

Identity / name	Contract	Description / details
ACT-REQUESTER Requester		Kind: person Description: A person who submits a request to the register.

Term

Identity / name	Contract	Description / details
VOC-REQUEST-CODE Request code		Definition: The exact text identifier of a retained request. Synonyms: none

Entities and validations

The entity holds request data and defines validity; it has no collaborators.

ENTITY ENT-REQUEST

Name Request

Meaning

A request asks the register to retain a quantity of a catalog code. The request is identified by its code.

Identity

Description

Two requests are the same when their codes are equal.

Fields: FLD-REQUEST-CODE

Fields

Identity / name	Contract	Description / details
FLD-REQUEST-CODE Request code	Type: text Cardinality: 1 Required: true	Description: A non-empty code chosen by the requester. Constraints: The code is preserved exactly; case is significant. Examples: Valid: REQ-001; req-001; Invalid:
FLD-REQUEST-QUANTITY Quantity	Type: integer Cardinality: 1 Required: true	Description: The number of units requested. Constraints: The value must be an integer greater than or equal to 1. Examples: Valid: 1; 2; Invalid: 0; 1.5

Relationships: none

States: none

Transitions: none

Validations

VLD-REQUEST-QUANTITY

Name Quantity is a positive integer Scope field

Statement

The quantity field is valid when its value is an integer greater than or equal to 1.

Inputs

Identity / name	Contract	Description / details
Quantity	Source: field	Meaning: The value supplied for FLD-REQUEST-QUANTITY.

Conditions

Identity / name	Contract	Description / details
C-VLD-REQ-INT The value is an integer.		Meaning: A fractional or textual value fails this condition.
C-VLD-REQ-MIN The value is at least 1.		Meaning: Zero and negative values fail this condition.

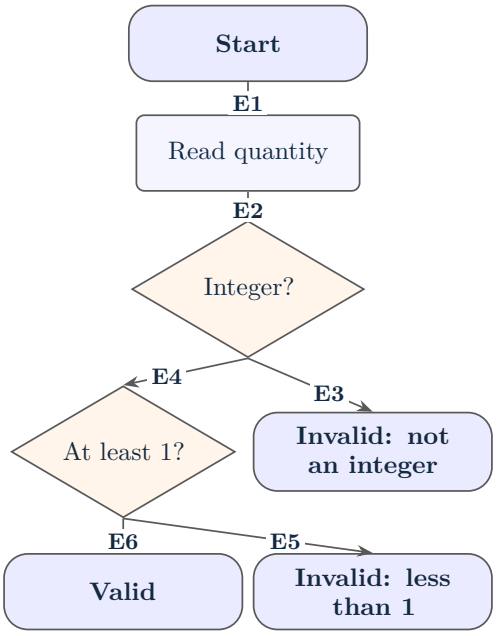
Results

Identity / name	Contract	Description / details
	Outcome: pass	Description: Both conditions hold; the field accepts the value.
	Outcome: fail	Description: At least one condition fails; the field rejects the value.

Flow GRAPH-VLD-REQUEST-QUANTITY

Quantity validation

GRAPH-VLD-REQUEST-QUANTITY



Key	Edge identity and label
E1	E-VLD-REQ-1
E2	E-VLD-REQ-2
E3	E-VLD-REQ-3 — no
E4	E-VLD-REQ-4 — yes
E5	E-VLD-REQ-5 — no
E6	E-VLD-REQ-6 — yes

VLD-REQUEST-CODE

Name	Request code is non-empty text	Scope	field

Statement

The code must be text containing at least one character. Preserve the code exactly; case is significant.

Inputs

Identity / name	Contract	Description / details
Code	Source: field	Meaning: FLD-REQUEST-CODE candidate.

Conditions

Identity / name	Contract	Description / details
C-VLD-REQ-CODE-NONEMPTY The code is text with at least one character.		Meaning: An empty or non-text code fails.

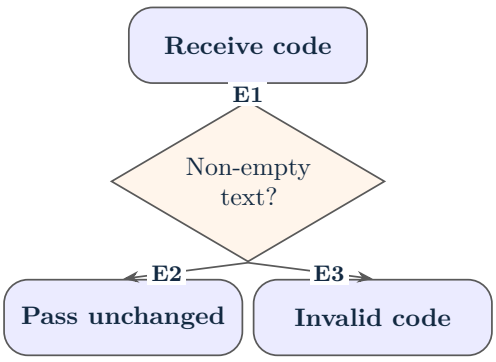
Results

Identity / name	Contract	Description / details
	Outcome: pass	Description: Accept unchanged.
	Outcome: fail	Description: Refuse before persistence.

Flow	GRAPH-VLD-REQUEST-CODE

Request code validity

GRAPH-VLD-REQUEST-CODE



Key	Edge identity and label
E1	E-VLD-REQ-CODE-1
E2	E-VLD-REQ-CODE-2 — yes
E3	E-VLD-REQ-CODE-3 — no

Examples

EX-REQUEST-VALID

Kind valid

Description

A request with a non-empty code and a positive quantity.

Data

```
{
"code": "REQ-001",
"quantity": 2
}
```

EX-REQUEST-INVALID

Kind invalid

Description

A request whose quantity is zero.

Data

```
{
"code": "REQ-001",
"quantity": 0
}
```

Assumptions: ASM-REQUEST-STORE-UNIQUE

Open issues: OI-REQUEST-LATENCY

Abstract tools

The tool exposes a capability and a contract, not an implementation.

TOOL TOL-REQUEST-STORE

Name Request store

Purpose

Retain request records durably. The contract does not choose a database, transport, or file layout.

Operations

OP-REQUEST-STORE

Name Store request

Purpose

Create one retained request record when its code is unused.

Inputs

Identity / name	Contract	Description / details
Request code	Type: text Required: true	Meaning: The code to retain.
Quantity	Type: integer Required: true	Meaning: An already validated positive integer.

Conditions

The code is not allocated to a retained request.

The store is available.

Inputs described as ready entities already satisfy canonical entity invariants. Do not duplicate those validations inside this tool. When constructing entities from raw retained records, use the canonical entity hydration rules; operation-specific external conditions remain this tool's responsibility.

Results

Identity / name	Contract	Description / details
Stored request	Type: REQUEST	Meaning: The retained request with its code and quantity. Guarantees: After a successful result, the record is readable by code in the same retained store.

Effects

One request record is added to the retained store.

Failures

Identity / name	Contract	Description / details
DUPLICATE_CODE		Meaning: The code is already allocated. Consumer distinction: The consumer must distinguish this from a successful store and from an unavailable store.
STORE_UNAVAILABLE		Meaning: The store did not confirm the write. Consumer distinction: Return write_outcome=not_applied only when the tool can guarantee no write; otherwise return write_outcome=unknown. Neither outcome is confirmation. An unknown outcome must not trigger an automatic retry.

Guarantees

A successful result confirms one retained record.

A duplicate-code failure leaves the existing record unchanged.

Use cases

Functional and non-functional requirements are declared in the case.

USE CASE UC-REQUEST-SUBMIT

Name Submit request

Objective

Retain one valid request and confirm the stored record.

Actors

Actor	ACT-REQUESTER	Permission	Permitted to submit requests.

Description

The requester supplies a code and quantity.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

The requester submits the request form.

Context

One request is submitted against the single request register.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-REQUESTER. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Identity / name	Contract	Description / details
Request code	Type: text Required: true	Meaning: The code that identifies the request.
Quantity	Type: integer Required: true	Meaning: The number of units requested.

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity ENT-REQUEST

Validations: VLD-REQUEST-QUANTITY, VLD-REQUEST-CODE

Purpose

Hold the supplied code and quantity and validate the quantity.

Tools

Tool	TOL-REQUEST-STORE	Operation	OP-REQUEST-STORE

Purpose

Retain the request after validation succeeds.

Calls: none

Outcomes

OUT-REQUEST-CONFIRMED

Kind success

Description

The request is retained and confirmed.

Effects

One new request record exists in the store.

OUT-REQUEST-INVALID

Kind refusal

Description

The code or quantity is invalid; the request is refused.

Effects

No store call and no retained record.

OUT-REQUEST-DENIED

Kind refusal

Description

The actor is absent or not permitted.

Effects

No validation, store call, or retained record.

OUT-REQUEST-STORE-FAILURE

Kind failure

Description

The store reports that it is unavailable.

Effects

No confirmed request is reported.

OUT-REQUEST-DUPLICATE

Kind refusal

Description

The request code is already retained.

Effects

This invocation leaves the existing request unchanged.

Functional requirements

RF-REQUEST-01

Statement

The case admits only an actor with the requester permission and refuses every other actor before any validation, store call, or effect.

Scope case

Applies to: UC-REQUEST-SUBMIT, INT-REQUEST-SUBMIT

Flow references: N-REQ-PERM, N-REQ-DENY, E-REQ-2

RF-REQUEST-02

Statement

The case applies VLD-REQUEST-QUANTITY to the supplied quantity and, when the validation reports fail, refuses the request with the invalid-data outcome and no store call.

Scope case

Applies to: UC-REQUEST-SUBMIT, INT-REQUEST-SUBMIT

Flow references: N-REQ-VALIDATE, N-REQ-QTYDEC, N-REQ-INVALID, E-REQ-4, E-REQ-5

RF-REQUEST-03

Statement

After a passing validation, the case calls OP-REQUEST-STORE with the code and quantity; a stored result produces the confirmation outcome.

Scope operation

Applies to: UC-REQUEST-SUBMIT, OP-REQUEST-STORE

Flow references: N-REQ-STORE, E-REQ-6, E-REQ-7

RF-REQUEST-04

Statement

When OP-REQUEST-STORE reports STORE_UNAVAILABLE, the case reports the store-unavailable failure and does not claim a confirmed request.

Preserve whether the tool guarantees not_applied or reports unknown. A lost confirmation is not evidence of no write; do not automatically retry.

Scope operation

Applies to: UC-REQUEST-SUBMIT, OP-REQUEST-STORE

Flow references: N-REQ-STOREERROR, E-REQ-8

RF-REQUEST-05

Statement

A confirmed request result contains the supplied code and quantity and identifies the new record.

Scope interaction

Applies to: UC-REQUEST-SUBMIT, INT-REQUEST-SUBMIT

Flow references: N-REQ-SUCCESS, E-REQ-7

RF-REQUEST-06

Statement

When OP-REQUEST-STORE reports DUPLICATE_CODE, refuse the duplicate and leave the existing request unchanged; do not report confirmation.

Scope operation

Applies to: UC-REQUEST-SUBMIT, OP-REQUEST-STORE

Flow references: N-REQ-DUPLICATE, E-REQ-DUPLICATE

RF-REQUEST-07

Statement

After authorization and before quantity validation, apply VLD-REQUEST-CODE. Refuse invalid code without storage; preserve valid code exactly.

Scope case

Applies to: UC-REQUEST-SUBMIT

Flow references: N-REQ-INVALID, N-REQ-CODE, N-REQ-CODE-VALID, E-REQ-CODE-2, E-REQ-CODE-3

Nonfunctional requirements

RNF-REQUEST-01

Scope interaction

Applies to: INT-REQUEST-SUBMIT

Statement

For inputs within the declared field limits and an available store, the boundary presents a confirmed or refused result within 2 seconds of the submit stimulus.

Verifiable criterion

The elapsed time from the submit stimulus to the visible result is at most 2 seconds.

Verification

Run 30 sequential submissions with valid and invalid quantities against the declared local configuration and record the result time for each.

Metric

Name	Visible result time	Unit	seconds
Target	2	Operator	<=

Measurement

Start

The submit stimulus is accepted by the boundary.

End

The result state is visible to the actor.

References: SRC-REQUEST-BRIEF

Applicability

Origin: SRC-REQUEST-BRIEF. This is a synthetic teaching obligation; any numeric target and scenario count are illustrative assumptions, not an accepted production SLA or a measurement made by this repository.

Flows

Main	GRAPH-REQUEST-SUBMIT

Subflows: none

Flow text

Start: the case receives the request and actor. If the actor is not permitted, refuse with no effects (RF-REQUEST-01).

Apply VLD-REQUEST-CODE before the quantity check. An invalid code is refused without storage (RF-REQUEST-07).

Apply VLD-REQUEST-QUANTITY (RF-REQUEST-02). A fail result refuses the request as invalid data.

On pass, call OP-REQUEST-STORE (RF-REQUEST-03). A stored result produces the confirmed outcome (RF-REQUEST-05). STORE_UNAVAILABLE produces the store-unavailable failure (RF-REQUEST-04).

DUPLICATE_CODE is the duplicate refusal with no change to the preexisting request (RF-REQUEST-06). A write failure retains its known-not-applied versus unknown distinction (RF-REQUEST-04).

Acceptance criteria

AC-REQUEST-01

Statement

A requester with a valid code and quantity of 2 receives a confirmed result; a later lookup of the code returns quantity 2.

Verifies: RF-REQUEST-02, RF-REQUEST-03, RF-REQUEST-05

AC-REQUEST-02

Statement

A quantity of 0 is refused as invalid data, and no record is created.

Verifies: RF-REQUEST-02

AC-REQUEST-03

Statement

An actor absent from the permitted set is refused before validation or store access.

Verifies: RF-REQUEST-01

AC-REQUEST-04

Statement

Submit the same code twice. The second attempt reports duplicate-code refusal and preserves the first retained quantity.

Verifies: RF-REQUEST-06

AC-REQUEST-05

Statement

If a store write has an unknown final outcome, return the failure without claiming success or no write and without automatic retry.

Verifies: RF-REQUEST-04

AC-REQUEST-06

Statement

An empty code is refused before quantity validation and store invocation.

Verifies: RF-REQUEST-07

AC-REQUEST-SUBMIT-QUALITY

Statement

In the declared didactic configuration, demonstrate the canonical quality criterion and its stated verification procedure; preserve its status as a teaching assumption, not a measured production claim.

Verifies: RNF-REQUEST-01

Traceability

Identity / name	Contract	Description / details
TRC-REQUEST-01	Source: RF-REQUEST-02	Targets: VLD-REQUEST-QUANTITY; AC-REQUEST-02 Note: The case applies the entity validation and states the refusal.
TRC-REQUEST-02	Source: RF-REQUEST-03	Targets: OP-REQUEST-STORE; AC-REQUEST-01 Note: The abstract tool contract carries the persistence guarantee.
TRC-REQUEST-SUBMIT-ORIGIN	Source: SRC-REQUEST-BRIEF	Targets: RF-REQUEST-01; RF-REQUEST-02; RF-REQUEST-03; RF-REQUEST-04; RF-REQUEST-05; RF-REQUEST-06; RF-REQUEST-07; RNF-REQUEST-01 Note: Versioned synthetic origin for these example obligations.

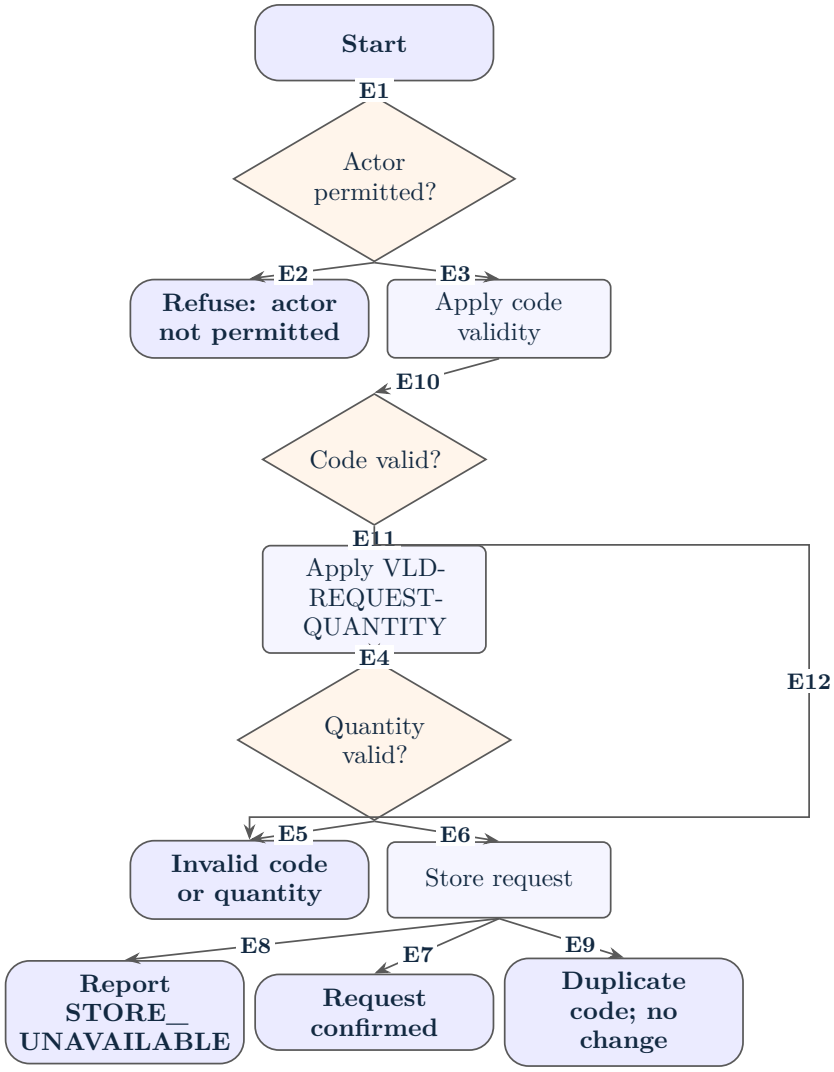
Assumptions: ASM-REQUEST-STORE-UNIQUE

Sources: SRC-REQUEST-BRIEF

Open issues: OI-REQUEST-LATENCY

Submit request

GRAPH-REQUEST-SUBMIT



Key	Edge identity and label
E1	E-REQ-1
E2	E-REQ-2 — no
E3	E-REQ-3 — yes
E4	E-REQ-4
E5	E-REQ-5 — fail
E6	E-REQ-6 — pass
E7	E-REQ-7 — stored
E8	E-REQ-8 — STORE_UNAVAILABLE
E9	E-REQ-DUPLICATE — DUPLICATE_CODE
E10	E-REQ-CODE-1
E11	E-REQ-CODE-2 — pass
E12	E-REQ-CODE-3 — fail

Interaction boundaries

The boundary maps the case inputs and results to a web form.

BOUNDARY BND-REQUEST-WEB

Name	Request register web form	Modality	web

Purpose

Expose the submit-request case through one form.

Origin classification: this channel is an illustrative solution proposal. Its form, command syntax, layout and encoding are outside the L1 conformance claim. The referenced cases define domain inputs and outcomes independently of this channel.

Audience

Requesters using the local register.

Interactions

INT-REQUEST-SUBMIT

Name	Submit request form

Description

A form with Request code, Quantity, and Submit request. The result area presents confirmation, invalid data, duplicate code, or store failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case	UC-REQUEST-SUBMIT

Trigger

The requester activates Submit request.

Input mapping

Source	form.code	Target	UC-REQUEST-SUBMIT.input.code

Meaning

Maps the Request code field to the case code input.

Source	form.quantity	Target	UC-REQUEST-SUBMIT.input.quantity

Meaning

Maps the Quantity field to the case quantity input.

Result mapping

Source	OUT-REQUEST-CONFIRMED	Target	result.confirmation

Meaning

The request is retained and confirmed.

Source	OUT-REQUEST-INVALID	Target	result.invalid_data

Meaning

The code or quantity is invalid; the request is refused.

Source	OUT-REQUEST-STORE-FAILURE	Target	result.store_unavailable

Meaning

The store reports that it is unavailable.

Source	OUT-REQUEST-DENIED	Target	result.out-request-denied

Meaning

The actor is absent or not permitted.

Source	OUT-REQUEST-DUPLICATE	Target	result.out-request-duplicate

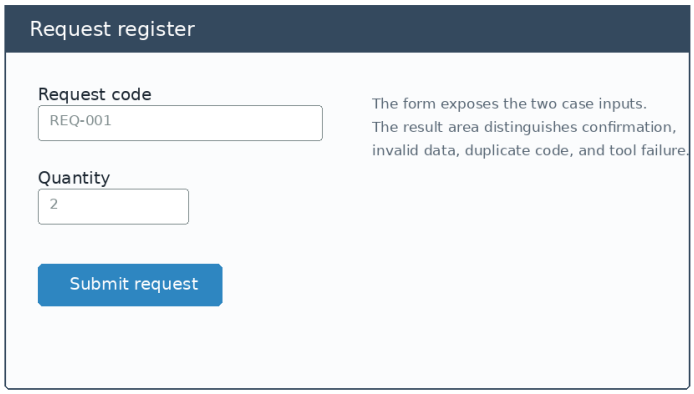
Meaning

The request code is already retained.

Requirement refs: RF-REQUEST-01, RF-REQUEST-02, RF-REQUEST-03, RF-REQUEST-04, RF-REQUEST-05, RF-REQUEST-06, RF-REQUEST-07

Resources: RES-REQUEST-WIRE, RES-REQUEST-FLOW

Examples



Wireframe with Request code, Quantity, and Submit request.

Alternative text:

A form with a Request code field, a Quantity field, and a Submit request button.

Assumptions: ASM-REQUEST-STORE-UNIQUE

Sources: SRC-REQUEST-BRIEF

Open issues: OI-REQUEST-LATENCY

Appendix: graphs

Identity / name	Contract	Description / details
GRAPH-VLD-REQUEST-QUANTITY	validation	VLD-REQUEST-QUANTITY; Figure
GRAPH-REQUEST-SUBMIT	use_case	UC-REQUEST-SUBMIT; Figure
GRAPH-VLD-REQUEST-CODE	validation	VLD-REQUEST-CODE; Figure

Graph `GRAPH-VLD-REQUEST-QUANTITY`

Quantity validation

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-VLD-REQ-START	Type: start	Label: Start Requirement refs: none
N-VLD-REQ-READ	Type: action	Label: Read quantity Requirement refs: none
N-VLD-REQ-TYPE	Type: decision	Label: Integer? Requirement refs: none
N-VLD-REQ-MIN	Type: decision	Label: At least 1? Requirement refs: none
N-VLD-REQ-PASS	Type: result	Label: Valid Requirement refs: none
N-VLD-REQ-FAIL-TYPE	Type: refusal	Label: Invalid: not an integer Requirement refs: none
N-VLD-REQ-FAIL-MIN	Type: refusal	Label: Invalid: less than 1 Requirement refs: none

Edges

Identity / name	Contract	Description / details
E-VLD-REQ-1	From: N-VLD-REQ-START To: N-VLD-REQ-READ	Requirement refs: none
E-VLD-REQ-2	From: N-VLD-REQ-READ To: N-VLD-REQ-TYPE	Requirement refs: none
E-VLD-REQ-3	From: N-VLD-REQ-TYPE To: N-VLD-REQ-FAIL-TYPE	Requirement refs: none Label: no
E-VLD-REQ-4	From: N-VLD-REQ-TYPE To: N-VLD-REQ-MIN	Requirement refs: none Label: yes
E-VLD-REQ-5	From: N-VLD-REQ-MIN To: N-VLD-REQ-FAIL-MIN	Requirement refs: none Label: no
E-VLD-REQ-6	From: N-VLD-REQ-MIN To: N-VLD-REQ-PASS	Requirement refs: none Label: yes

Topology of GRAPH-VLD-REQUEST-QUANTITY:
Nodes:

- N-VLD-REQ-START (start)
- N-VLD-REQ-READ (action)
- N-VLD-REQ-TYPE (decision)
- N-VLD-REQ-MIN (decision)
- N-VLD-REQ-PASS (result)
- N-VLD-REQ-FAIL-TYPE (refusal)
- N-VLD-REQ-FAIL-MIN (refusal)

Edges:
- E-VLD-REQ-1: N-VLD-REQ-START -> N-VLD-REQ-READ
- E-VLD-REQ-2: N-VLD-REQ-READ -> N-VLD-REQ-TYPE
- E-VLD-REQ-3: N-VLD-REQ-TYPE -> N-VLD-REQ-FAIL-TYPE
- E-VLD-REQ-4: N-VLD-REQ-TYPE -> N-VLD-REQ-MIN
- E-VLD-REQ-5: N-VLD-REQ-MIN -> N-VLD-REQ-FAIL-MIN
- E-VLD-REQ-6: N-VLD-REQ-MIN -> N-VLD-REQ-PASS

Figure: GRAPH-VLD-REQUEST-QUANTITY

Graph `GRAPH-REQUEST-SUBMIT`

Submit request

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-REQ-START	Type: start	Label: Start Requirement refs: none
N-REQ-PERM	Type: decision	Label: Actor permitted? Requirement refs: RF-REQUEST-01
N-REQ-DENY	Type: refusal	Label: Refuse: actor not permitted Requirement refs: RF-REQUEST-01
N-REQ-VALIDATE	Type: validation	Label: Apply VLD-REQUEST-QUANTITY Requirement refs: RF-REQUEST-02 Call: Kind: validation; Identifier: VLD-REQUEST-QUANTITY
N-REQ-QTYDEC	Type: decision	Label: Quantity valid? Requirement refs: RF-REQUEST-02
N-REQ-INVALID	Type: refusal	Label: Invalid code or quantity Requirement refs: RF-REQUEST-02; RF-REQUEST-07
N-REQ-STORE	Type: operation	Label: Store request Requirement refs: RF-REQUEST-03 Call: Kind: operation; Identifier: OP-REQUEST-STORE
N-REQ-STOREERROR	Type: failure	Label: Report STORE_UNAVAILABLE Requirement refs: RF-REQUEST-04
N-REQ-SUCCESS	Type: result	Label: Request confirmed Requirement refs: RF-REQUEST-05
N-REQ-DUPLICATE	Type: refusal	Label: Duplicate code; no change Requirement refs: RF-REQUEST-06
N-REQ-CODE	Type: validation	Label: Apply code validity Requirement refs: RF-REQUEST-07 Call: Kind: validation; Identifier: VLD-REQUEST-CODE

Identity / name	Contract	Description / details
N-REQ-CODE-VALID	Type: decision	Label: Code valid? Requirement refs: RF-REQUEST-07

Edges

Identity / name	Contract	Description / details
E-REQ-1	From: N-REQ-START To: N-REQ-PERM	Requirement refs: none
E-REQ-2	From: N-REQ-PERM To: N-REQ-DENY	Requirement refs: RF-REQUEST-01 Label: no
E-REQ-3	From: N-REQ-PERM To: N-REQ-CODE	Requirement refs: none Label: yes
E-REQ-4	From: N-REQ-VALIDATE To: N-REQ-QTYDEC	Requirement refs: RF-REQUEST-02
E-REQ-5	From: N-REQ-QTYDEC To: N-REQ-INVALID	Requirement refs: RF-REQUEST-02 Label: fail
E-REQ-6	From: N-REQ-QTYDEC To: N-REQ-STORE	Requirement refs: RF-REQUEST-03 Label: pass
E-REQ-7	From: N-REQ-STORE To: N-REQ-SUCCESS	Requirement refs: RF-REQUEST-03; RF-REQUEST-05 Label: stored
E-REQ-8	From: N-REQ-STORE To: N-REQ-STOREERROR	Requirement refs: RF-REQUEST-04 Label: STORE_UNAVAILABLE
E-REQ-DUPLICATE	From: N-REQ-STORE To: N-REQ-DUPLICATE	Requirement refs: RF-REQUEST-06 Label: DUPLICATE_CODE
E-REQ-CODE-1	From: N-REQ-CODE To: N-REQ-CODE-VALID	Requirement refs: none
E-REQ-CODE-2	From: N-REQ-CODE-VALID To: N-REQ-VALIDATE	Requirement refs: RF-REQUEST-07 Label: pass
E-REQ-CODE-3	From: N-REQ-CODE-VALID To: N-REQ-INVALID	Requirement refs: RF-REQUEST-07 Label: fail

Topology of GRAPH-REQUEST-SUBMIT:
Nodes:
- N-REQ-START (start)
- N-REQ-PERM (decision)
- N-REQ-DENY (refusal)
- N-REQ-VALIDATE (validation)
- N-REQ-QTYDEC (decision)
- N-REQ-INVALID (refusal)

- N-REQ-STORE (operation)
- N-REQ-STOREERROR (failure)
- N-REQ-SUCCESS (result)
- N-REQ-DUPLICATE (refusal)
- N-REQ-CODE (validation)
- N-REQ-CODE-VALID (decision)

Edges:

- E-REQ-1: N-REQ-START -> N-REQ-PERM
- E-REQ-2: N-REQ-PERM -> N-REQ-DENY
- E-REQ-3: N-REQ-PERM -> N-REQ-CODE
- E-REQ-4: N-REQ-VALIDATE -> N-REQ-QTYDEC
- E-REQ-5: N-REQ-QTYDEC -> N-REQ-INVALID
- E-REQ-6: N-REQ-QTYDEC -> N-REQ-STORE
- E-REQ-7: N-REQ-STORE -> N-REQ-SUCCESS
- E-REQ-8: N-REQ-STORE -> N-REQ-STOREERROR
- E-REQ-DUPLICATE: N-REQ-STORE -> N-REQ-DUPLICATE
- E-REQ-CODE-1: N-REQ-CODE -> N-REQ-CODE-VALID
- E-REQ-CODE-2: N-REQ-CODE-VALID -> N-REQ-VALIDATE
- E-REQ-CODE-3: N-REQ-CODE-VALID -> N-REQ-INVALID

Figure: GRAPH-REQUEST-SUBMIT

Graph `GRAPH-VLD-REQUEST-CODE`

Request code validity

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-VLD-REQ-CODE-START	Type: start	Label: Receive code Requirement refs: none
N-VLD-REQ-CODE-CHECK	Type: decision	Label: Non-empty text? Requirement refs: none
N-VLD-REQ-CODE-PASS	Type: result	Label: Pass unchanged Requirement refs: none
N-VLD-REQ-CODE-FAIL	Type: refusal	Label: Invalid code Requirement refs: none

Edges

Identity / name	Contract	Description / details
E-VLD-REQ-CODE-1	From: N-VLD-REQ-CODE-START To: N-VLD-REQ-CODE-CHECK	Requirement refs: none
E-VLD-REQ-CODE-2	From: N-VLD-REQ-CODE-CHECK To: N-VLD-REQ-CODE-PASS	Requirement refs: none Label: yes

Identity / name	Contract	Description / details
E-VLD-REQ-CODE-3	From: N-VLD-REQ-CODE-CHECK To: N-VLD-REQ-CODE-FAIL	Requirement refs: none Label: no

Topology of GRAPH-VLD-REQUEST-CODE:
Nodes:
- N-VLD-REQ-CODE-START (start)
- N-VLD-REQ-CODE-CHECK (decision)
- N-VLD-REQ-CODE-PASS (result)
- N-VLD-REQ-CODE-FAIL (refusal)

Edges:
- E-VLD-REQ-CODE-1: N-VLD-REQ-CODE-START -> N-VLD-REQ-CODE-CHECK
- E-VLD-REQ-CODE-2: N-VLD-REQ-CODE-CHECK -> N-VLD-REQ-CODE-PASS
- E-VLD-REQ-CODE-3: N-VLD-REQ-CODE-CHECK -> N-VLD-REQ-CODE-FAIL

Figure: GRAPH-VLD-REQUEST-CODE

Appendix: resources

Resource `RES-REQUEST-WIRE`

Path: `assets/request-form.svg`

Media type: image/svg+xml

SHA-256: `f5a97fb7cb43af9681b2083fe1deebe38bb3eda24adb947c3314317fc8a24de9`

Drawing source

Path	assets/request-form.svg	Revision	1
Tool	scripts/build_examples.py		

Caption

Request form wireframe: Request code, Quantity, and Submit request.

Alternative text

A form with a Request code field, a Quantity field, and a Submit request button. A result area reports confirmation, invalid data, duplicate code, or tool failure.

Text present in the image

- Request code
- Quantity
- Submit request

Variants

Identity / name	Contract	Description / details
		Path: assets/request-form.png Media type: image/png Sha256: 5dd57bf8bccc9bcd9b68e1be1ea50163777caf93d83f6fbfec1a099eb1d84347 Size: 22242 Relation: publication Derived from: assets/request-form.svg Producer: scripts/build_examples.py

Resource RES-REQUEST-FLOW

Path: assets/request-flow.svg

Media type: image/svg+xml

SHA-256: 71f3a97b667a88041cc5812da8c8e7e1ddaced2f9c835e56c2c7d6d1b4252206

Drawing source

Path	assets/request-flow.svg	Revision	1286331986e92aa7 0d90981f34da7aaa 1dc45b09fa69106b 4717745a2a095c0a
Tool	scripts/build_examples.py		

Derived from graph: GRAPH-REQUEST-SUBMIT

Caption

Submission flow with authorization, code and quantity validation, persistence, duplicate refusal and explicit write failure. The edge key identifies every structured edge.

Alternative text

Every node and edge of GRAPH-REQUEST-SUBMIT, with wrapped labels and an edge key. The text and graph records provide the complete accessible description.

Text present in the image

- Start
- Actor permitted?
- Refuse: actor not permitted
- Apply VLD-REQUEST-QUANTITY
- Quantity valid?
- Invalid code or quantity
- Store request
- Report STORE_UNAVAILABLE
- Request confirmed
- Duplicate code; no change
- Apply code validity
- Code valid?
- no
- yes
- fail
- pass
- stored
- STORE_UNAVAILABLE
- DUPLICATE_CODE
- pass
- fail

Variants

Identity / name	Contract	Description / details
		Path: assets/request-flow.png Media type: image/png Sha256: 53545111afdda2bcf5b113667ef14dc2 233ca3f7e5e730749d3f50473ed81f8b Size: 73547 Relation: publication Derived from: assets/request-flow.svg Producer: scripts/build_examples.py

Appendix: assumptions

ASM-REQUEST-STORE-UNIQUE

Statement

A future implementation of the request store can enforce code uniqueness atomically.

Status open

Affected: TOL-REQUEST-STORE, UC-REQUEST-SUBMIT

Appendix: decisions

DEC-MIN-TOOL-ABSTRACTION

Title Keep the store abstract

Context

The example needs persistence but must not choose a technology.

Decision

Define OP-REQUEST-STORE as a tool contract only.

Alternatives

Specify SQLite tables and SQL statements.

Consequences

Implementations may choose storage, but must satisfy the contract.

Status accepted

Refs: TOL-REQUEST-STORE

Appendix: open issues

OI-REQUEST-LATENCY

Title Confirm the latency target

Description

The 2-second target is a verifiable example target; a production adoption must confirm it with stakeholders.

Status open

Refs: RNF-REQUEST-01

Appendix: sources

SRC-REQUEST-BRIEF

Title	Illustrative request-register brief	Uri	null
Revision	2	Accessed	null

Note

Internal teaching source; not a commissioned requirement.

The authorization policy, interaction sketches, and quality targets are didactic choices, not observed business facts.

Appendix: history

HIST-REQUEST-01

Version	1	Date	2026-10-25

Description

Initial reference example for the first Atlas Specifications edition, Alpha-v0.1.0. This synthetic example demonstrates the model and its representations; it does not assert stakeholder approval or execution of the described business application.

HIST-MIGRATION-STABLE

Version	2	Date	2026-10-25

Description

Explicit editorial migration from Atlas Alpha-v0.1.0 revision 1 to Atlas Stable-v1.0.0 revision 2. Model/profile/envelope and producer identities change; existing concept IDs, domain contracts, authored history and resources are retained. This is not format conversion, a Needs-to-L1 derivation, or new stakeholder approval.

Derived summary: 5 sections; 1 actors; 1 entities; 1 tools; 1 use cases; 1 boundaries; 3 graphs; 2 resources.